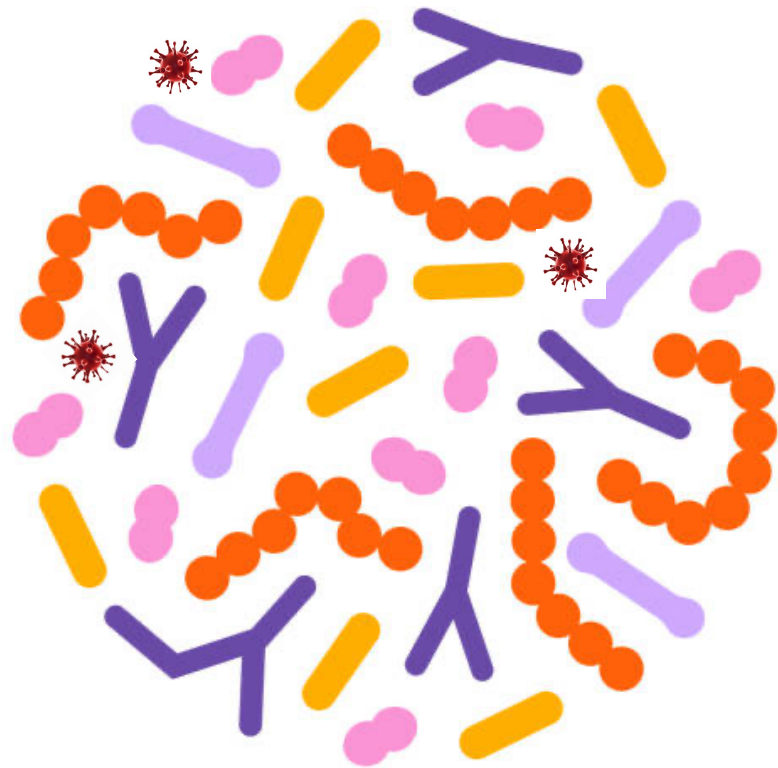




Processing of shotgun metagenomics samples

- nf-core/taxprofiler together with Metaphlan v4 database
- Genome-level marker gene based taxonomic classifier
- Data Science platform
- DTU – Biosustain
- Spring Data-type Courses
- 11th June 2025

Metagenomic insights into the prokaryotic communities of heavy metal-contaminated hypersaline soils

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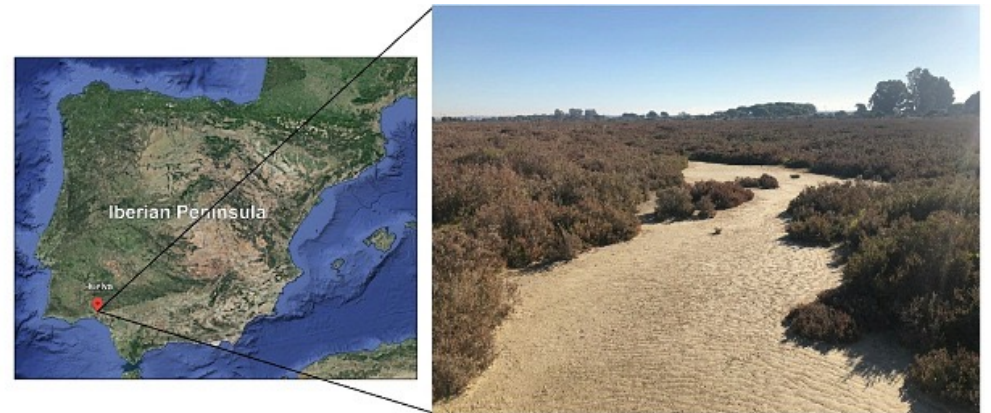
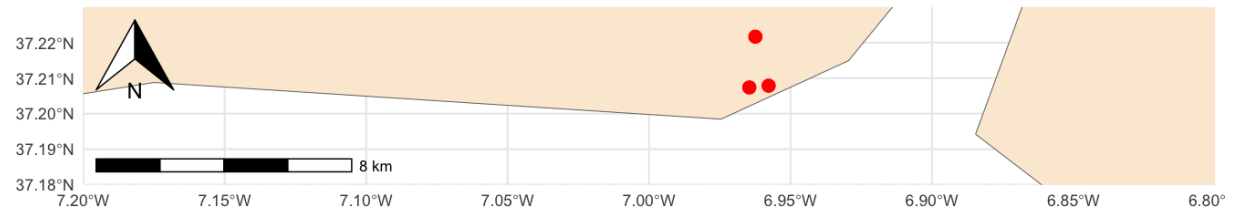
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- Saline soils and their microbial communities have recently been studied in response to ongoing desertification of agricultural soils caused by anthropogenic impacts and climate change.
- Here they describe the prokaryotic microbiota of hypersaline soils in the Odiel Saltmarshes Natural Area of Southwest Spain.
- This region has been strongly affected by mining and industrial activity and feature high levels of certain heavy metals.
- 18 shotgun metagenomes sequenced through Illumina NovaSeq from three different areas in 2020 and 2021.

Sampling sites

- Three sampling sites (1, 2 and 3) were selected in order to obtain a representative subset of the prokaryotic community of the soils from the Odiel Saltmarshes Natural Area



Sample table with some metadata:

- Sampling date
- Area
- Coordinates
- Sample names

Sampling	Date	Area	Samples	Coordinates
M2	July 2020	1	M2_1A	37°12'26.6"N 6°57'52.5"W
			M2_1B	
			M2_1C	
		2	M2_2A	37°12'28.4"N 6°57'27.9"W
			M2_2B	
			M2_2C	
		3	M2_3A	37°13'18.0"N 6°57'44.8"W
			M2_3B	
			M2_3C	
M3	June 2021	1	M3_1A	37°12'26.6"N 6°57'52.5"W
			M3_1B	
			M3_1C	
		2	M3_2A	37°12'28.4"N 6°57'27.9"W
			M3_2B	
			M3_2C	
		3	M3_3A	37°13'18.0"N 6°57'44.8"W
			M3_3B	
			M3_3C	

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A00180 → instrument name

10 → run id

H7NK5DMXX → flowcell id

1 → flowcell lane

1101 → tile number within the flow cell line

16821 → x-coordinate of the cluster within the tile

1000 → y-coordinate of the cluster within the tile

SPACE

1 → member of the pair (if paired-end)

N → N if read is not filtered, Y if it is filtered

0 → when none of the control bit are on

GGACTT, GTCGTTTCG → barcodes are used for demultiplexing and assign reads to the correct sample



Four lines for each read:

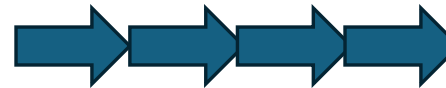
1. identifier (starting with @)
2. Read sequence
3. a line starting with + (sometimes id gets repeated)
4. quality scores per nucleotide (Phred score)

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTCC  
TG CAGCAGCTAATGAGGAACCACTTCTCCTCCAGCGCTCTAAATACTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAACTG  
+  
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF  
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTCC  
TG CAGAAGAAGAAAGCACAA GTATT TACGCCTATCCTTCATATTTCCGCAAGTA ACTATCTCGGTTTCATATCGAGATTATATA GAATCT  
+  
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF - FFFFFFFFFF  
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTCC  
TG CAGGAAGTTATGCAGGGCATCTGTATTATTAATAGACGACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG  
+  
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-F-FFFFFFF  
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTCC  
TG CAGAGTACATCAACAAAAGAACTA A CTGCCCTACCGCAAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCAGTCAATATAAGG  
+  
8F888FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFFFF-FFFFFFFFFFFFFFFFF-FFFFFFFFF
```

Repeating steps too many times? – find yourself a pipeline!

```
@A00180:10:H7NK5DMXX:1:1101:16821:1000 1:N:0:GGACTT+GTCGTTTCG
TGCAGCAGCTAATGAGGAACCACTTCTCCCTCCAGCGCTCTAAATACCTCAGAACAATAGGATCATCATAATAATCCCCTAGTCTGAAGTG
+
8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19090:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGAAGAAGAAAGCACAAGTATTACGCCTATCCTTCATATTTCCGCAAGGTAAGTATCTCGGTTTCATATCGAGATTATATAGAATCT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:19325:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGGAAGTTATGCAGGGGATCCTGTATTATTAATAAGAGCACCTACTCTTCATAGATTAGGTATACAGGCGTTCCAACCTATTTTAGTGG
+
FF-88F8FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00180:10:H7NK5DMXX:1:1101:30897:1016 1:N:0:GGACTT+GTCGTTTCG
TGCAGAGTACATCAACAAAAGAAACCTAACTGCCCTACCGGCAACCGGTAGAGTACCCTTCCCCAAAAGTATTACTCCAGTCAATATAAGG
+
8F88FFF-FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
```

Series of steps in
an specific order



Abundance table for
downstream analysis

OTU	Sample 1	Sample 2	Sample 3
A	0.60	0	0.70
B	0.24	0.20	0.10
C	0.10	0	0
D	0.05	0	0
E	0.01	0	0
F	0	0.80	0.20
Total	1.0	1.0	1.0

- Different steps to run manually in an order (QC, host removal, mapping to a database, etc...)
- Workflow that contains all the processes needed and have them chained

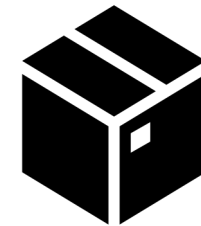
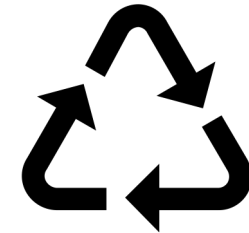


Nextflow is a language to build workflows and a software to orchestrate workflows execution.

- Strong community behind with various bioinformaticians joining forces to build each of the pipelines
- Support for other technologies

Nextflow's core features

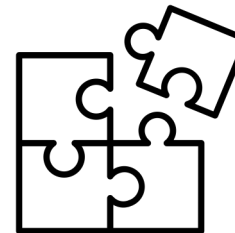
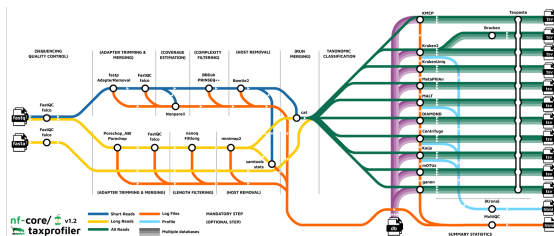
- **Reproducible** between runs
 - integration with code management tools
 - all packages downloaded, organized in containers, and control over computing environment
- **Portable** between systems
 - you can write the code in your laptop and can run everywhere (HPC, cloud)
 - works with most of computing environments
- **Scalable**
 - it can run for 10 samples on your laptop or thousands in the cloud
- **Integration** of existing tools, systems, and industry standards





A global community effort to collect a curated set of open-source analysis pipelines built using Nextflow.
<https://nf-co.re/>

<https://www.nature.com/articles/s41587-020-0439-x>



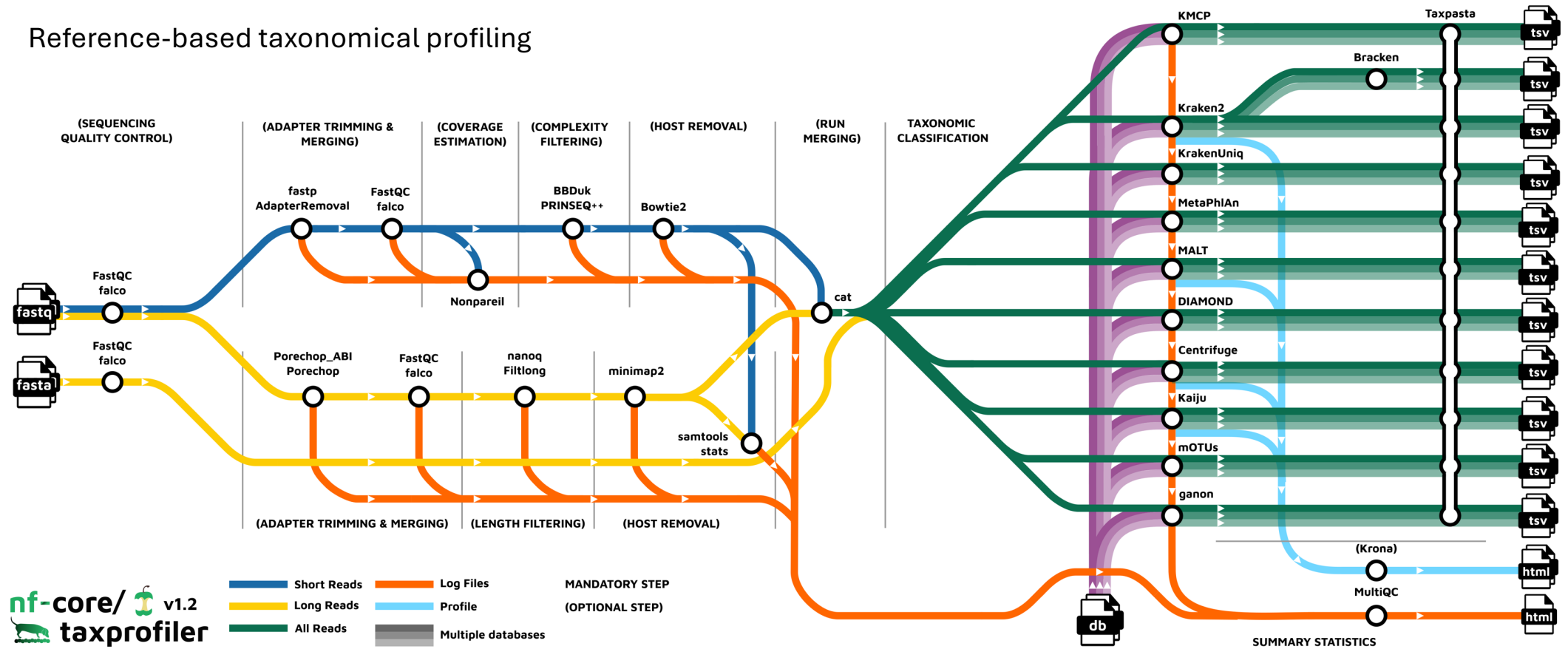
120 pipelines available
(8th June 2025)
to process and analyse
many different data types

Modules: 1,527
reusable
components that
can be integrated
into pipelines

Subworkflows: 83 pre-
assembled
combinations of
modules aimed at
streamlining commonly
used workflows

nf-core/taxprofiler

Reference-based taxonomical profiling



Read QC (FastQC)

- [FastQC](#) gives general quality metrics about the sequenced reads.
- It provides information about the quality score distribution across your reads, per base sequence content, adapter contamination and overrepresented sequences.

in work/<hexadecimal code>/command.sh

```
[ ! -f SRR28149497_SRR_1.fastq.gz ] && ln -sf
SRR28149497_1.fastq.gz SRR28149497_SRR_1.fastq.gz
[ ! -f SRR28149497_SRR_2.fastq.gz ] && ln -sf
SRR28149497_2.fastq.gz SRR28149497_SRR_2.fastq.gz
fastp \
  --in1 SRR28149497_SRR_1.fastq.gz \
  --in2 SRR28149497_SRR_2.fastq.gz \
  --out1 SRR28149497_SRR_1.fastp.fastq.gz \
  --out2 SRR28149497_SRR_2.fastp.fastq.gz \
  --json SRR28149497_SRR.fastp.json \
  --html SRR28149497_SRR.fastp.html \
  \
  \
  \
  --thread 6 \
  --detect_adapter_for_pe \
  --detect_adapter_for_pe --length_required 15 \
  2> >(tee SRR28149497_SRR.fastp.log >&2)
```

Adapter trimming for Illumina data ([fastp](#))

[fastp](#) is a FASTQ pre-processing tool for:

- quality control
- trimmming of adapters
- quality filtering

```
[ ! -f SRR28149497_SRR_1.fastq.gz ] && ln -sf SRR28149497_1.fastq.gz SRR28149497_SRR_1.fastq.gz  
[ ! -f SRR28149497_SRR_2.fastq.gz ] && ln -sf SRR28149497_2.fastq.gz SRR28149497_SRR_2.fastq.gz
```

```
fastp \  
    --in1 SRR28149497_SRR_1.fastq.gz \  
    --in2 SRR28149497_SRR_2.fastq.gz \  
    --out1 SRR28149497_SRR_1.fastp.fastq.gz \  
    --out2 SRR28149497_SRR_2.fastp.fastq.gz \  
    --json SRR28149497_SRR.fastp.json \  
    --html SRR28149497_SRR.fastp.html \  
    \  
    \  
    \  
    --thread 6 \  
    --detect_adapter_for_pe \  
    --detect_adapter_for_pe --length_required 15 \  
2> >(tee SRR28149497_SRR.fastp.log >&2)
```

Genome-level marker gene based taxonomic classifier ([MetaPhlAn](#))

[MetaPhlAn](#) is a computational tool for profiling the composition of microbial communities (Bacteria, Archaea and Eukaryotes) from metagenomic shotgun sequencing data (i.e. not 16S) with species-level resolution via marker genes.

```
BT2_DB=`find -L "metaphlan_db" -name "*rev.1.bt2*" -exec dirname {} \`;`  
BT2_DB_INDEX=`find -L metaphlan_db -name "*rev.1.bt2*" | sed 's/\.\rev.1.bt2.*$//' | sed  
's/.*\V//'`
```

```
metaphlan \  
  --nproc 6 \  
  --input_type fastq \  
  SRR28149499_SRR_1.fastp.fastq.gz,SRR28149499_SRR_2.fastp.fastq.gz \  
  \  
  --bowtie2out SRR28149499_SRR_db1.metaphlan.bowtie2out.txt \  
  --bowtie2db $BT2_DB \  
  --index $BT2_DB_INDEX \  
  --output_file SRR28149499_SRR_db1.metaphlan_profile.txt
```

Aggregate report describing results and QC from the whole pipeline ([MultiQC](#))

- [MultiQC](#) is a visualization tool that generates an HTML report summarising all samples in your project
- Most of the pipeline QC results are visualised in the report
- Further statistics from all software that can be reported in multiQC are available in the report

```
multiqc \  
  --force \  
  --config multiqc_config.yml \  
  --cl-config 'custom_logo: "nf-core-  
taxprofiler_logo_custom_light.png"' \  
  .
```

Command line to run the pipeline

```
nextflow run nf-core/taxprofiler -r 1.2.3 -profile test,docker --outdir processed_data
```